

SET C (Try other sets)

1. $\vec{N} = (0, 0, 1)$

2. $\vec{V} = (9, 3, 6) - (1, 1, 0) = (8, 2, 6)$
 $= \left(\frac{8}{7}, \frac{2}{7}, \frac{6}{7}\right)$

3. $A = I_a k_a = I_a k_a$
 $= 0.2 \times 0.4 \times (0.6, 0.1, 0.3)$
 $= (0.048, 0.008, 0.024)$

4. $I_p = 0.7, \vec{L} = (-1, -2, 6) - (1, 1, 0)$
 $= (-2, -3, 6)$
 $= \left(-\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right)$ [Normalizing]

a. $\vec{R} = 2(\vec{L} \cdot \vec{N}) \cdot \vec{N} - \vec{L}$
 $= \frac{12}{7}(0, 0, 1) - \left(-\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right)$
 $= \left(\frac{2}{7}, \frac{3}{7}, \frac{6}{7}\right)$

b. $D = 0.7 \times 0.6 \times (0.6, 0.1, 0.3)$
 $\times \max((\vec{L} \cdot \vec{N}), 0)$
 $= (0.252, 0.042, 0.126) \times \frac{6}{7}$
 $= (0.216, 0.036, 0.108)$

c. $S = 0.7 \times 0.5 \times \max((\vec{R} \cdot \vec{V}), 0)^2$
 $= 0.35 \times \left(\frac{6}{49} + \frac{36}{49} + \frac{6}{49}\right)^2$
 $= 0.35 \times \left(\frac{48}{49}\right)^2 = 0.336$

d. $d = \sqrt{(-1-1)^2 + (-2-1)^2 + (6-0)^2}$
 $= \sqrt{4 + 9 + 36} = 7$

$r = 10$

$f_{\text{att}} = 1 - \left(\frac{d}{r}\right)^2 = 1 - \frac{49}{100}$
 $= 0.51$

$I = A + f_{\text{att}}(D + S)$
 $= (0.048, 0.008, 0.024) + 0.51 \times$
 $((0.216, 0.036, 0.108) + (0.336, 0.336, 0.336))$
 $= (0.33, 0.2, 0.25)$

5. Phong shading. Other shadings misses the effect of spotlight.